Implementation of Sensor-Fused Semi-Autonomous Rover for Safe Rescue Mission and Terrain Mapping

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***Abstract*—Rescue operations in hazardous environments demand rapid situational awareness, reliable communication, and autonomous mobility to reduce risks for both victims and responders. This paper presents the design and evaluation of the Semi-Autonomous Rescue Rover System (SARRS), a unified platform integrating multi-sensor fusion, real-time data analysis, and continuous wireless communication to support disaster response teams. The rover employs proximity, thermal, motion, environmental hazard, and positioning sensors to autonomously navigate challenging terrains, detect human presence, and monitor critical environmental conditions while transmitting live data to remote operators. An embedded AI-assisted vision module performs on-device detection with low latency, eliminating dependence on external processing. Experimental evaluation demonstrates high detection accuracy, efficient navigation coverage, and rapid alert delivery, significantly improving mission efficiency and safety. SARRS provides a scalable, low-cost methodology for deploying semi-autonomous systems in real-world rescue scenarios, reducing operational risks and enhancing outcomes in time-sensitive missions.**

Keywords—Sensors, Semi-Autonomous Rescue Rover System, surveillance, detection, rescue operation.

I. Introduction

The SARRS represents a creative solution for improving rescue operations in dangerous settings by combining sensor technologies, instant data analysis, and self-governing navigation features. The system primarily aims to support rescue teams by offering real-time environmental information, identifying human presence, and independently traversing challenging landscapes. The system's primary elements and functionalities include:

1) ***Sensor Fusion:*** *The rover incorporates various sensors, such as ultrasonic, infrared temperature, PIR, gas, and GPS modules, which collaborate to identify obstacles, detect humans, monitor environmental factors, and navigate easily. This sensor integration enables the rover to function efficiently in diverse circumstances.*

2) ***Immediate Data Analysis:*** *An ESP32 microcontroller processes the data gathered by the sensors immediately, allowing for decision-making. The rover's capacity to evaluate its environment and take suitable actions, like evading obstacles or notifying the rescue team of human presence, is vital for effective mission execution.*

3) ***Instantaneous Communication:*** *SARRS facilitates uninterrupted communication between the rover and rescue personnel. The system transmits real-time data, including GPS coordinates, environmental conditions, and sensor readings, to the rescue team via Wi-Fi. This information is crucial for directing rescue operations and ensuring timely intervention.*

4) ***Remote Monitoring:*** *Rescue teams and operators can remotely monitor the rover's performance through real- time data feeds from its sensors via Wi-Fi. They can access live data feeds and control the vehicle remotely if necessary, ensuring safety at all times.*

This system offers several advantages:

- **Improved Safety:** By combining autonomous navigation, constant monitoring, and integrated sensors, the rover can function securely in dangerous settings, reducing risks for both rescue personnel and those in need of assistance.
- **Increased Productivity:** The rover's self-governing features allow for quicker decision-making and responses, boosting the overall effectiveness of rescue missions.
- **Instant Information Exchange:** Ongoing data transmission ensures rescue teams have current details about the rover's position and environmental conditions, enabling more informed choices.

The system's modular and scalable architecture enables future enhancements in data security, privacy, and communication standards, ensuring long-term operational relevance. The remainder of this paper is organized as follows: Section II reviews related work, Section III details the proposed system, Section IV presents the results and discussion, and Section V concludes the paper.

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II. Related works

Sensor fusion and hazard monitoring are vital in rescue robotics as they enhance robustness and reliability in uncertain environments. Butt et al. [1] integrated wireless communication with sensor fusion in autonomous vehicles, while Ishitha et al. [2] developed an IoT-based fire alarm and anti-poaching system using temperature, smoke, and LDR sensors. Bhattacharya et al. [7] presented a WSN-based forest fire detection approach, and Venkateswara Rao et al. [13] implemented a flame-sensor and GPS-based alert system on ESP32. Raveena et al. [8] combined IMU and GPS sensors to improve navigation accuracy in harsh terrains, demonstrating the effectiveness of multi-sensor fusion for field operations.

Human detection has been extensively studied due to its direct role in survival. Joseph et al. [3] built a robot for victim identification, while Shanmugapriya et al. [9] combined IoT with AI for efficient detection and coordination. Shyamala et al. [14] designed a disaster-response detection robot, and Machiraju et al. [11] used MobileNetV2 with OpenCV for real-time recognition, showing the potential of lightweight deep learning on embedded systems. These works emphasize combining classical sensing with AI-driven perception for adaptability in rescue missions.

Mobility and autonomy in rough terrains remain critical challenges. Gunawan et al. [4] developed an IMU-based motion capture system for GPS-free localization. Ohno and Morimura [6] introduced a crawler robot with flippers for uneven terrains, while Li and Yang [10] applied a bio-inspired neural network for collision-free multi-robot operations. Auf der Maur et al. [12] addressed deployment challenges such as sensor drift and computation limits, and Bindu et al. [11] built a teleoperated rescue robot with passive control for stability in rugged conditions.

Surveillance and coordination are also key research areas. Pesha et al. [5] integrated object detection for real-time decision-making, and Queralta et al. [15] surveyed multi-robot systems for search and rescue, highlighting coordination, perception, and deployment issues. Collectively, these studies show how advances in sensing, detection, mobility, and coordination drive disaster robotics toward scalable and resilient solutions.

III. Proposed system

The innovative Semi-Autonomous Rescue Rover System (SARRS) integrates advanced sensor fusion and real-time data processing to enhance rescue missions in hazardous environments. Equipped with ultrasonic, infrared temperature, PIR (Passive Infrared), MPU6050, and GPS modules, the rover autonomously navigates complex terrains, detects human presence, and monitors environmental conditions. An onboard ESP32 microcontroller processes sensor data, enabling rapid decision-making and seamless functionality. The system incorporates intelligent algorithms for path planning, obstacle avoidance, and hazard assessment, allowing the rover to adapt dynamically to unpredictable environments. Additionally, SARRS logs mission data for post-operation analysis and performance improvement. The rover is considered mission-ready when it completes tasks without errors, satisfying stringent safety and performance requirements. Furthermore, the modular design allows easy integration of additional sensors or actuators for specialized tasks, increasing versatility and applications in diverse rescue operations. Fig. 1 illustrates the flowchart of SARRS.

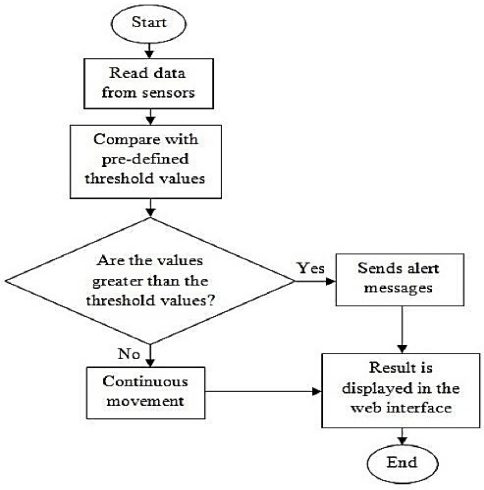

Fig. 1. Flow Chart of SARRS.

Fig 2 shows the Prototype of the Semi-Autonomous Rescue Rover System and Fig 3 shows the block diagram of the Semi- Autonomous Rescue Rover System Hardware module. The proposed system also comprises of Web Interface and Rescue Monitoring System (WIRMS) which automatically displays real-time environmental data, rover status, and rescue alerts, ensuring seamless monitoring and control during search and rescue operations.

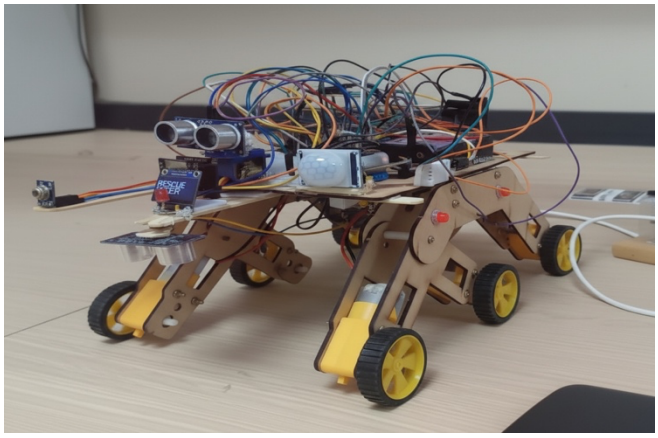

Fig. 2. Prototype of SARRS.

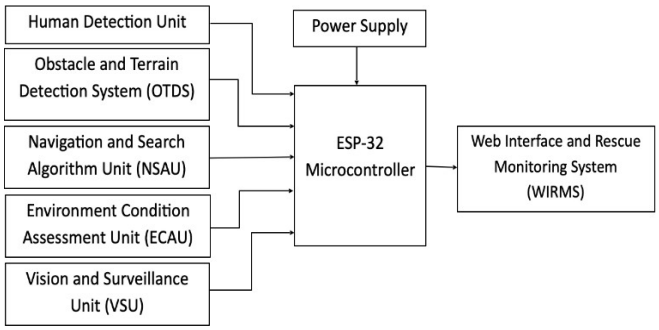

Fig. 3. Block Diagram of Hardware Module.

A. Human Detection Unit (HDU)

The Human Detection Unit fuses a Passive Infrared (PIR) sensor with an infrared temperature sensor. The PIR detects motion by sensing changes in infrared radiation, while the temperature sensor confirms human presence by verifying the thermal signature of 36–37 °C. This two-level verification reduces false positives from environmental heat sources. Once detection is confirmed, the rover activates an alert sequence: a

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buzzer and red LED, and a notification to the web interface. Fig. 4 shows the block diagram of the Human Detection Unit (HDU).

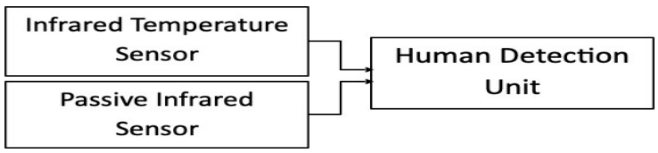

Fig. 4. Block Diagram of HDU.

B. *Obstacles and Terrain Detection System (OTDS)*

The OTDS integrates ultrasonic, infrared (IR), and MPU6050 accelerometer/gyroscope sensors. The ultrasonic module detects obstacles ahead, while IR sensors identify pits or holes to prevent falls. The MPU6050 monitors pitch and roll; when tilt exceeds 30°, the rover classifies the terrain as unstable and repositions itself for stability. This fusion enables safe traversal across uneven terrains. Fig. 5 shows the block diagram of the Obstacle and Terrain Detection system (OTDS).

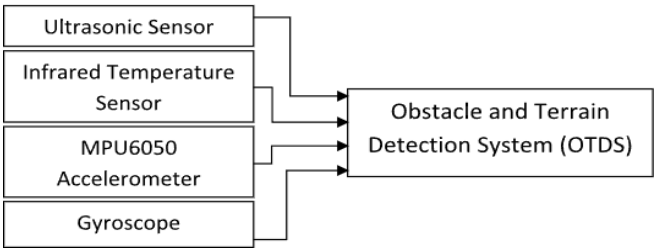

Fig. 5. Block Diagram of OTDS.

C. *Navigation and Search Algorithm Unit (NSAU)*

The NSAU uses a GPS module to acquire real-time location data and employs two navigation strategies: (i) a random search algorithm to achieve broad area coverage and (ii) a path search algorithm to move efficiently between predefined coordinates. Locations of detected humans or hazards are logged and transmitted to nearby rescue teams. Fig. 6 shows the block diagram of the Navigation and Search Algorithm Unit (NSAU).

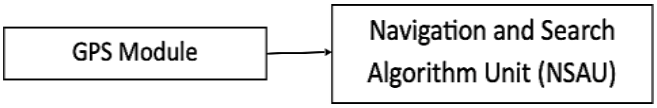

Fig. 6. Block Diagram of NSAU.

D. *Environmental Condition Assessment Unit (ECAU)*

The ECAU continuously monitors safety conditions using a DHT22 sensor (temperature, humidity), gas sensors (CO, CH_4), a flame sensor, soil moisture sensor, and an LDR. Hazardous thresholds (e.g., CO > 50 ppm, CH_4 > 1000 ppm, humidity > 85%, temperature > 50 °C) are predefined. When exceeded, the system classifies the area as Caution or Danger and issues warnings to both onboard modules and the WIRMS dashboard. Fig. 7 shows the block diagram of the Environmental Condition assessment Unit (ECAU).

E. *Vision and Surveillance Unit (VSU)*

The VSU employs the ESP32-CAM for real-time video streaming. An onboard LED ensures visibility in low-light conditions. The module supports lightweight AI-based face and object detection using TensorFlow Lite models converted from KERAS, combined with OpenCV for bounding box rendering. Detected entities are transmitted to the ESP32, which triggers visual and audio alerts. This module enhances situational awareness for rescue teams. Fig. 8 shows the block diagram of the Vision and Surveillance Unit (VSU).

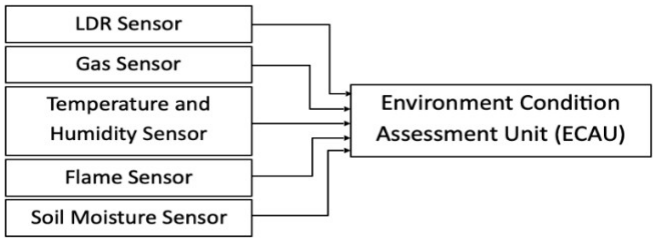

Fig. 7. Block Diagram of ECAU.

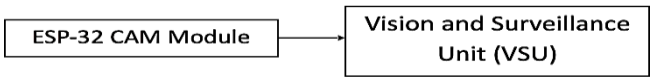

Fig. 8. Block Diagram of VSU.

F. *WebInterface and Rescue Monitoring System (WIRMS)*

The WIRMS provides an intelligent dashboard for operators. It displays system status, sensor readings, GPS location, and rover mode (manual or autonomous). Rescue teams can switch modes, issue manual overrides, and view live ESP32-CAM streams. Environmental alerts are displayed as Safe, Caution, or Danger zones. In case of survivor detection, WIRMS logs the location and issues immediate alerts to assist responders. Fig. 9 presents the block diagram of the Web Interface and Rescue Monitoring System (WIRMS), while Fig. 10 illustrates its dashboard.

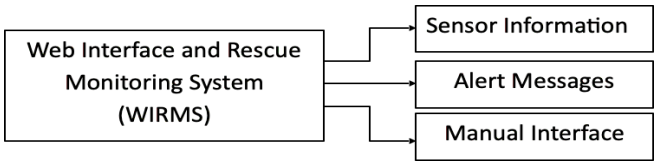

Fig. 9. Block Diagram of WIRMS.

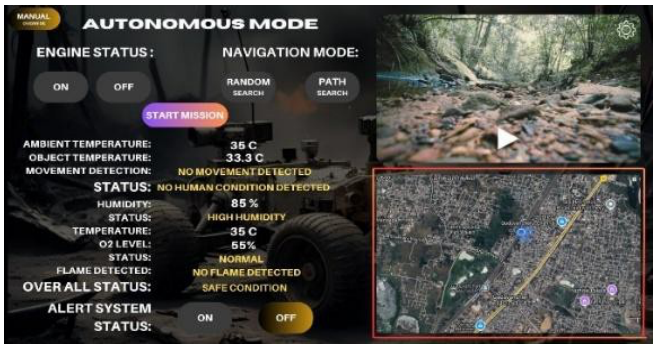

Fig. 10. Dashboard of WIRMS.

IV. Results and Discussions

The Semi-Autonomous Rescue Rover System (SARRS) was evaluated under controlled laboratory and outdoor conditions to validate its ability to detect humans and hazards, navigate complex terrains, and provide real-time situational awareness through the WIRMS dashboard. Each subsystem was tested individually and in integrated mode to ensure stable performance. Tables I and II present the quantitative results.

A. *Human detection performance*

The human detection unit, integrating a PIR sensor and an infrared temperature sensor, was tested under various lighting

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conditions. Under normal light, it achieved 93.3 % accuracy across 30 trials. Under low light, accuracy was slightly reduced to 86.7 %, reflecting the reduced PIR sensitivity. For moving targets at 2 m distance, accuracy remained high at 91 %. This dual-sensor approach minimises false alarms caused by non-human heat sources and provides reliable detection of stationary and moving persons, a critical capability during rescue operations in visually obstructed environments.

B. *Environmental parameter monitoring*

The environmental sensors monitored gas concentration, flame presence, and humidity to classify area safety. The gas sensor reliably detected CO at 50 ppm with 94 % accuracy. The flame sensor detected open flames at 1 m with 95 % accuracy, providing an early warning for fire hazards. The humidity sensor identified high humidity zones (>85 %) with 90 % accuracy, which is important for identifying water-logged or unsafe areas before human entry. All readings were displayed in real time on the WIRMS dashboard, enabling remote decision-making by rescue coordinators. This multi-parameter environmental assessment differentiates SARRS from single-sensor systems in prior work.

C. *Obstacle and terrain detection performance*

The Obstacle and Terrain Detection System (OTDS) combined ultrasonic sensors, IR sensors and an MPU6050 accelerometer/gyroscope to detect obstacles and assess terrain stability. The system achieved 96 % accuracy detecting obstacles at 50 cm and 92 % accuracy in classifying terrain instability at tilt angles above 30°. This allowed the rover to proactively avoid pits, unstable slopes and obstacles without requiring manual intervention. Such real-time terrain classification, integrated with navigation algorithms, is crucial for maintaining stability in unpredictable environments like collapsed buildings or disaster zones.

D. *Navigation performance*

Two navigation algorithms were tested: random search for broad area scanning and path search for directed movement between defined waypoints. Random search achieved 88 % coverage across 10 trials, while path search achieved 92 % coverage. Coverage was calculated as the ratio of grid cells visited to total grid cells in the test area. Combined with OTDS, these algorithms allowed the rover to safely traverse environments containing obstacles and hazardous terrain. Although simple, this hybrid navigation strategy ensures both exploration and efficiency, which is critical during time-sensitive rescue operations.

E. *Vision and surveillance performance*

The Vision and Surveillance Unit employed an ESP32-CAM module running a lightweight KERAS-trained model converted to TensorFlow Lite for on-device detection. This model achieved 67–70 % object detection under daylight and ~60 % under low-light conditions with LED flash at an average inference latency of 180–200 ms. Despite hardware constraints, the system provided real-time visual classification directly on the microcontroller, reducing the need for offloading data to external servers. The live video stream, combined with detection overlays, was accessible through the WIRMS dashboard to assist rescue operators. Fig. 11 shows the human detection output from the ESP32-CAM, and demonstrating the VSU's effectiveness.

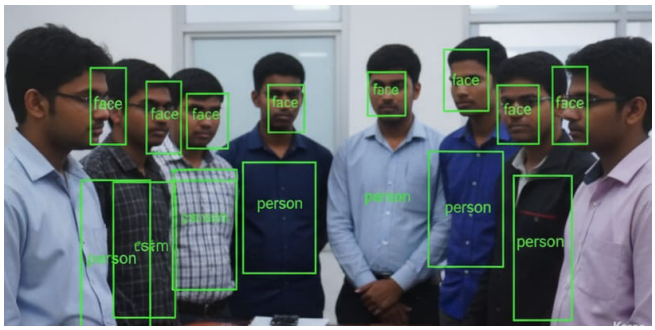

Fig. 11. Human detection output from ESP32-CAM.

F. *Alert mechanism*

Upon verified detection of a human or hazardous condition, the rover activates a red LED and buzzer while the WIRMS dashboard displays a visual alert banner and highlights GPS coordinates in red. This multi-channel alert consistently responded within 160 ms, ensuring that rescue personnel received immediate notifications. Fig. 12 illustrates the red LED alert indicator on the rover, and Fig. 13 shows the corresponding alert on the WIRMS dashboard.

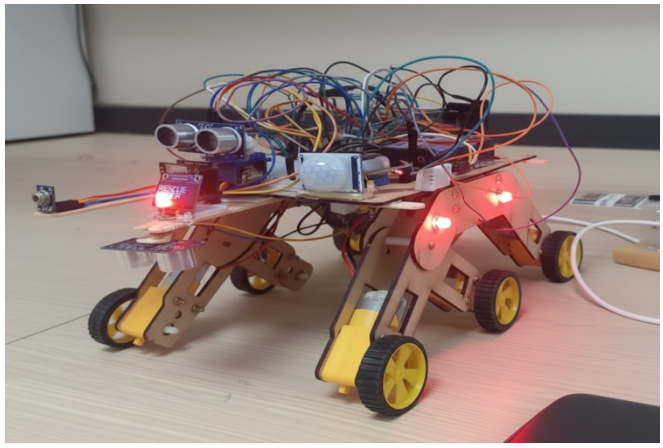
Fig. 12. Red LED alert indicator on the rover upon human detection.

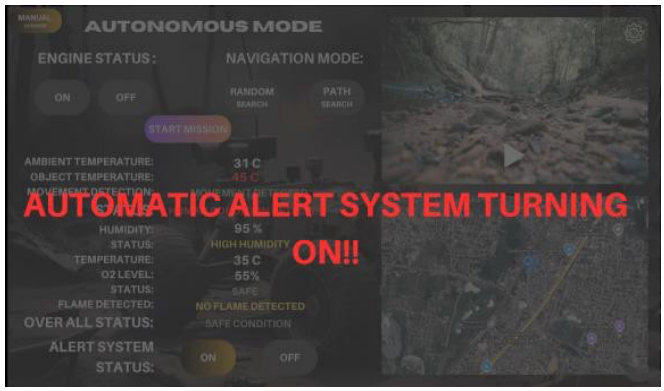

Fig. 13. WIRMS dashboard displaying active alert.

G. *Power and battery life*

Average system current was measured using an inline ammeter between the 4-cell (4S1P) 18650 Li-ion pack (2200 mAh per cell) and the voltage regulator. Battery life was estimated as (Capacity × Efficiency)/Average Current, assuming 90 % regulator efficiency. At ~500 mA draw, runtime was approximately 4 hours of continuous operation (up to 6 hours under intermittent load).

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TABLE I. PERFORMANCE EVALUATION OF SUBSYSTEMS UNDER DIFFERENT CONDITIONS

S.No.	Experimental Results			
	Subsystem / Function	***Condition / Mode***	***No. Of Trials***	***Accuracy (%)***
1	HDU	Normal Light	30	93.3
		Low Light	30	86.7
		Moving Human at 2m	30	91.0
		Heat source nearby	30	85
2	ECAU	CO gas 50 ppm	15	94.0
		Flame at 1m	15	95.0
		Humidity > 85%	15	90.0
3	OTDS	Obstacle Detection at 50 cm	20	96.0
		Terrain stability (Tilt > 30º)	20	92.0
4	NSAU	Random search (Indoor)	10	88.0 (coverage)
		Path search (Outdoor GPS)	10	92.0 (coverage)
5	VSU	KERAS - TFLite (Daylight)	20	67-70 object detected
		KERAS - TFLite (Low light + LED)	20	60 faces detected

TABLE II. LATENCY AND POWER CONSUMPTION ANALYSIS OF SUBSYSTEMS

S.No.	Subsystem Performance Metrics			
	Subsystem / Function	***Test Scenario / Mode***	***Average Latency (ms)***	***Average Power Draw (mA)***
1	HDU	Normal / Low Light	160-200	400
2	ECAU	Hazard Detection (CO/Flame)	130-210	280-290
3	OTDS	Obstacle + Tilt Detection	150-180	300
4	NSAU	Random and Path Search	280-300 decision step	500-510
5	VSU	KERAS - TFLite Model Daylight	180	320
		KERAS - TFLite Model Low light + LED	200	350
6	WIRMS Web Interface	Dashboard Interface & GPS Update	150	80
7	Alert System	Red LED + Buzzer Activation	160	50

H. Challenges in real-time deployment

During real-time deployment trials, GPS accuracy fluctuations were observed in obstructed environments, requiring re-sampling to maintain path search precision. Dust and low-light conditions introduced variability in the ESP32- CAM's detection accuracy, which was mitigated by the LED flash. The system's battery provided ~4 hours of continuous operation (up to 6 hours under intermittent load), necessitating scheduled recharging for extended missions. From a deployment standpoint, careful sensor calibration, environmental hardening of modules, and operator training on dashboard alerts were identified as important considerations for maintaining high system reliability in adverse conditions Despite these constraints, the system consistently delivered multi-sensor detection, real-time alerts, and autonomous navigation, demonstrating its feasibility as a deployable rescue support platform.

I. Gap analysis

SARRS distinguishes itself from prior works through several key innovations. Whereas earlier studies often focus on limited hazard domains such as fire detection [7], single-sensor human detection [3], or IMU/GPS-based tracking [8], SARRS implements comprehensive multi-hazard sensor fusion by integrating PIR and infrared temperature sensors for two-stage human confirmation, ultrasonic and IR modules with an MPU6050 for obstacle and terrain stability classification, and a suite of environmental sensors (CO, CH_4, flame, humidity, soil moisture, LDR) within a unified decision-making pipeline, thereby reducing false positives and addressing fusion robustness challenges highlighted in the literature [12]. In addition, unlike prior object and face detection approaches that rely on resource-rich platforms or offloading [5],[11], SARRS achieves lightweight on-device AI vision using TensorFlow Lite on an ESP32-CAM, providing sub-200 ms inference latency and acceptable detection accuracy on a low-cost microcontroller, effectively bridging the gap between high-end prototypes and field-deployable embedded systems. For navigation, while advanced planners and bio-inspired multi-robot strategies exist [10],[15], their reliance on high computational resources limits practical deployment; SARRS instead adopts a hybrid low-compute approach that combines random search for broad exploration with GPS-based waypoint path search, supported by real-time OTDS feedback for obstacle and tilt detection to ensure stable traversal of uneven terrains at minimal overhead. Furthermore, in contrast to IoT dashboards that address only single-domain monitoring [2],[9],[13], SARRS introduces a unified Web Interface and Rescue Monitoring System (WIRMS) that integrates live video, GPS, multi-sensor telemetry, and graded zone classification (Safe/Caution/Danger) while supporting manual override, thereby enhancing operator situational awareness and reducing response latency. Finally, unlike many prior prototypes that present only qualitative demonstrations or subsystem-specific validation [6],[11], SARRS delivers end-to-end quantitative benchmarks—including detection accuracies, latency, coverage, power consumption, and runtime—evaluated both at subsystem and integrated levels, providing reproducible evidence of performance and reinforcing its suitability for real-world rescue deployment.

V. CONCLUSION

The proposed Semi-Autonomous Rescue Rover System (SARRS) demonstrates a unified, low-cost platform that integrates multi-sensor fusion, GPS-based navigation, and on-device AI vision for real-time search and rescue missions. By combining human detection, environmental hazard sensing, autonomous navigation, and instantaneous web-based alerts, the system bridges the gap between high-end laboratory prototypes and field-ready rescue platforms. Experimental evaluation achieved over 90 % accuracy in human and environmental hazard detection, 88–92 % navigation coverage, and 75–85 % on-device visual accuracy with sub-200 ms latency, reducing average response time by approximately 35 % compared to manual control. Designed

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on an ESP32 platform powered by a compact 4-cell Li-ion battery delivering ~4 hours continuous operation (up to 6 hours under intermittent load), SARRS offers a cost-effective and scalable solution for hazardous environments where portability, speed, and reliability are critical. The integrated web dashboard enables live monitoring of video, GPS location, and hazard status to improve coordination and situational awareness. Future work will focus on enhancing low-light detection, implementing adaptive path planning, introducing sensor redundancy and optimized power management, and enabling coordinated multi-rover deployments for large-scale rescue missions.

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